

REVIEW

Molecular Mechanisms Governing Vascular Function in Heart Failure

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ABSTRACT: Heart failure is a leading cause of morbidity and mortality worldwide. Emerging evidence points to vascular dysfunction as a major contributor to the development and progression of this grievous syndrome. Functional interplay between the myocardium and vasculature involves endothelial cells, vascular smooth muscle cells, and pericytes, governing tissue perfusion, vascular tone, inflammation, and metabolism. Vascular cells play a central role in maintaining cardiovascular homeostasis by sensing hemodynamic and inflammatory stimuli and releasing vasoactive and paracrine factors. Dysregulation of endothelium-derived mediators such as NO, endothelin-1, and endothelium-derived hyperpolarizing factors, along with vascular oxidative stress, inflammation, and vascular remodeling (eg, disturbances in angiogenesis and microvascular rarefaction) impair myocardial perfusion and can accelerate heart failure progression. Here, we summarize key molecular mechanisms linking vascular dysfunction to heart failure, emphasizing endothelial signaling, oxidative stress, inflammation, and angiogenic remodeling.

Key Words: coronary artery disease ■ endothelial cells ■ heart failure ■ inflammation ■ perfusion

Heart failure (HF), defined as an impairment in the heart's ability to pump blood sufficient to meet the body's metabolic needs, is a leading health concern worldwide.¹ The vascular system comprises an extensive and highly organized network of vessels distributed throughout the body and mediates nutrient delivery, gas exchange, metabolic waste removal, immune and inflammatory responses, hemostasis, and the regulation of blood pressure, tissue perfusion, and thermoregulation.² The heart and vascular system work as a hemodynamic unit, ensuring adequate systemic and regional blood flow. Disruption of mechanical homeostasis within this coupled system, resulting from altered shear stress, cyclic strain, or pressure overload, represents a critical determinant in the pathogenesis of both vascular and cardiac dysfunction.^{3,4}

Alterations in vascular function have been observed in preclinical models of HF. Studies employing genetic and pharmacological manipulation of specific vascular cell populations have afforded mechanistic insights into how vascular dysfunction, both systemic and coronary, contributes to cardiac impairment. Elucidating the molecular imbalances and intercellular crosstalk among vascular cell types such as endothelial cells (ECs),

vascular smooth muscle (VSM) cells, and pericytes, and their interactions with other cardiovascular cell populations, such as cardiomyocytes and cardiac fibroblasts, is essential for a comprehensive understanding of HF pathogenesis.

Given the broad scope of vascular involvement in HF, this review focuses on molecular mechanisms governing coronary microvascular remodeling and their contribution to HF progression, with emphasis on evidence derived from preclinical models.

CORONARY VASCULAR SYSTEM: AN OVERVIEW

During embryonic development, heart formation is accompanied by the development of the coronary vascular system through processes such as vasculogenesis (the de novo formation of vessels from progenitor ECs) and angiogenesis (sprouting from preexisting vessels). These processes are orchestrated by an array of molecular signaling pathways and growth factors, including FGF (fibroblast growth factor), the Hedgehog family, VEGFs (vascular endothelial growth factors) [family and

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Nonstandard Abbreviations and Acronyms

αSMA	smooth muscle actin alpha
Angpt2	angiopoietin-2
CAD	coronary artery disease
CMD	coronary microvascular disease
CSE	cystathionine γ-lyase
CYB5R3	cytochrome b5 reductase 3
CYP2J2	cytochrome P450 2J2
DPP-4	dipeptidyl peptidase-4
DUSP4	dual-specificity phosphatase 4
EC	endothelial cell
EDHF	endothelium-derived hyperpolarizing factor
EndMT	endothelial-to-mesenchymal transition
eNOS	endothelial NO synthase
ErbB	erythroblastic leukemia viral oncogene homolog
ERG	ETS-related gene
ESAM	EC adhesion molecule
ET-1	endothelin-1
ETA	endothelin receptor type A
ETB	endothelin receptor type B
ETS	E26 transformation-specific
FGF	fibroblast growth factor
FKBP1	FK506-binding protein-like
Fli1	friend leukemia integration 1
FoxO1	forkhead box O1
FSTL1	follicle-stimulating-like protein 1
HF	heart failure
HFD	high-fat diet
HFpEF	heart failure with preserved ejection fraction
HFrEF	heart failure with reduced ejection fraction
HIF-1	hypoxia-inducible factor 1
I/R	ischemia/reperfusion
ICAM-1	intercellular adhesion molecule 1
IGFBP4	insulin-like growth factor-binding protein 4
IL	interleukin
JAM	junctional adhesion molecule
Jph2	juxtaaphilin-2
LepR	Leptin receptor
LOX	lysyl oxidase
Metnrl	Meteorin-like
MI	myocardial infarction
mTOR	mammalian target of rapamycin
NOS	NO synthase
nNOS	neuronal NOS

Nrg1	neuregulin-1
PKC	protein kinase C
ROCK	Rho-associated protein kinase
ROS	reactive oxygen species
RV	right ventricular
S1pr1	sphingosine 1 phosphate receptor 1
sGC	soluble guanylate cyclase
SphK1	sphingosine kinase 1
sPLA2	secretory phospholipase A2
Tβ4	thymosin β4
TAC	thoracic aortic constriction
TGF-β	transforming growth factor β
TNF-α	tumor necrosis factor-α
VCAM-1	vascular cell adhesion molecule 1
VE-cadherin	vascular endothelial cadherin
VEGF	vascular endothelial growth factor
VEGFR2	vascular endothelial growth factor receptor 2
VSM	vascular smooth muscle
WT	wild-type

receptors)] and their receptors and ligands, neuropilins, TGF-β (transforming growth factor-β) receptors and ligands, and the Notch signaling pathway, which collectively promote development of the coronary vascular bed.⁵ Disruption of these processes can impair cardiac development and function, with downstream impact on the entire body.^{6,7}

The coronary vasculature comprises epicardial vessels, including coronary arteries with a diameter >500 μm that conduct blood flow with minimal resistance, and the coronary microvasculature, which includes prearterioles, arterioles, capillaries, and venules ranging from ≈100 to 500 μm. The microvasculature contributes to cardiac perfusion by regulating vascular tone, promoting angiogenesis that enables oxygenation, nutrient supply, and delivery of reparative factors to metabolically active or injured tissue, and participating in coagulation and inflammatory responses.^{2,8}

Acute dysregulation of coronary vascular mechanisms impairs vascular tone, causing an imbalance in endothelium-derived vasoactive agents, such as NO and ET-1 (endothelin-1). Chronic dysregulation further promotes vascular remodeling, perpetuating cardiac dysfunction over time.^{2,8}

CORONARY ARTERY AND MICROVASCULAR DYSFUNCTION IN HF

The progression of HF is classified into stages ranging from A to D. Stage A is defined by the presence of risk factors, stage B by structural changes in the heart, stage

Highlights

- Heart failure is tightly linked to coronary vascular dysfunction. Beyond the fact that the heart and vasculature operate as a coupled hemodynamic unit, alterations in molecular mechanisms primarily in the microcirculation but also at the systemic level are associated with cardiac remodeling and dysfunction.
- The regulation and modulation of endothelial-derived factors, both in their synthesis and in their responses, whether acting locally or remotely, govern cardiac homeostasis and contribute to myocardial dysfunction.
- Oxidative stress participates in a feed-forward loop that damages both blood vessels and the heart. Genetic or pharmacological modulation of oxidative stress can attenuate both vascular and myocardial injury and dysfunction.
- The vascular system, particularly endothelial cells, represents a first line of response in inflammation. Its role is broad and involves molecular changes and signaling pathways that modify the vascular barrier, rendering it more permissive. Also, the vascular system promotes cytokine production and responses, allowing immune cell infiltration that contributes to heart failure development and progression.
- In the setting of pathological cardiac stress, early proangiogenic responses are associated with microvascular remodeling and capillary rarefaction, culminating in impaired tissue perfusion. This process is influenced by crosstalk among myriad cell types within the myocardium and is regulated by key molecular mediators.

C by both structural and functional alterations, and stage D represents the final stage of HF, characterized by severe dysfunction and significant impairment in quality of life.⁹ Where the evidence allows, we revisit this staging framework throughout this review to indicate with which phase of HF progression a given mechanism is most strongly associated, and we propose a parallelism between HF progression, coronary vascular dysfunction phenotype, and the associated molecular mechanisms (Figure 1).

Based on ejection fraction, HF is classified as HF with reduced ejection fraction (HFrEF; ejection fraction $\leq 40\%$), marked by systolic dysfunction and associated with risk factors mainly involving ischemic events such as myocardial infarction (MI), or nonischemic events, such as hypertension. HF with preserved ejection fraction (HFpEF; ejection fraction $\geq 50\%$) is marked by diastolic dysfunction and comorbidities, such as obesity, hypertension, diabetes, atrial fibrillation, and kidney disease.⁹

Dysfunction in the coronary or epicardial arteries, referred to as coronary artery disease (CAD), or in the microvascular bed, referred to as coronary microvascular disease (CMD), have been associated with HF. CAD

results from a reduction in coronary artery diameter, which can promote HF related to ischemic heart disease, such as MI.¹⁰ Although atherosclerosis is currently considered the main cause of CAD, the underlying mechanisms are believed to be more complex and remain to be fully elucidated.¹⁰ Like hypertension, which affects the systemic circulation, CAD is a modifiable risk factor for HF, relevant to stages A and B.¹¹ A clinical study reported that the presence of CAD was associated with all-cause mortality in patients with HFrEF.¹² Pharmacological intervention in patients with HFrEF and CAD has been shown to reduce hospitalization due to HF-related complications.¹⁰

CMD is characterized by structural and functional remodeling of the microvascular bed, driving ischemia in the presence or absence of CAD.¹³ CMD begins with an imbalance of the vasoactive agents that regulate vascular tone, promoting vasoconstriction through endothelium-dependent or nonendothelium-dependent mechanisms.¹⁴ Endothelium-derived agents include NO, primarily responsible for vasodilation of the epicardial arteries, and EDHFs (endothelium-derived hyperpolarizing factors), primarily responsible for vasodilation of microvessels. ET-1 is the main endothelium-derived vasoconstrictor.¹⁴ Non-endothelium-dependent regulation of vascular tone relies on the autonomic nervous system, where an imbalance between parasympathetic and sympathetic activity promotes vasoconstriction.¹⁴ This functional imbalance is followed by structural remodeling, including hypertrophic inward remodeling (intimal thickening, smooth muscle cell proliferation) with vessel occlusion and reduced luminal diameter, perivascular fibrosis of prearterioles and arterioles, and reduced capillary density and diameter, driving capillary constriction.^{14,15}

At the molecular level, overproduction of reactive oxygen species (ROS) triggers uncoupling of eNOS (endothelial NO synthase), increasing production of reactive nitrogen species and promoting oxidative stress and activation of proinflammatory pathways, including cytokine release, adhesion molecule expression, platelet and leukocyte adhesion, leukocyte infiltration, and impaired barrier function.¹⁴

CMD is present in $\approx 66\%$ of patients with HFrEF¹⁶ and 72% of patients with HFpEF.^{17,18} Whereas patients with HFrEF have a functional CMD characterized by lower microvascular resistance and higher absolute coronary flow at rest, patients with HFpEF manifest higher microvascular resistance and lower absolute coronary flow at rest.¹⁸

PRECLINICAL MODELS OF HF AND CORONARY MICROVASCULAR REMODELING

HF progression has been recapitulated in numerous preclinical models. These models reproduce key features of

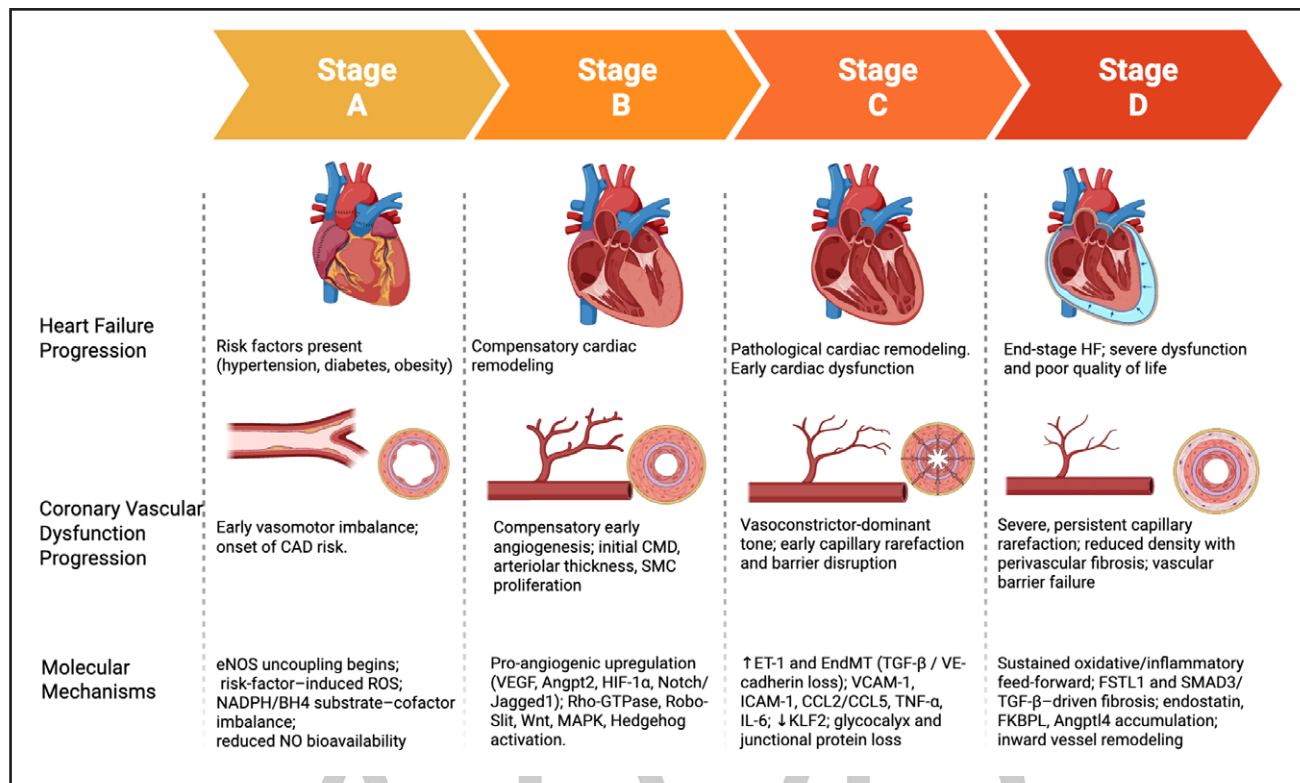


Figure 1. Integrative overview of coronary vascular mechanisms across the heart failure (HF) spectrum.

Stage-wise progression of coronary phenotypes across American College of Cardiology/American Heart Association HF stages A–D: stage A manifests early functional vasomotor imbalance driven by NO/EDHFs (endothelium-derived hyperpolarizing factors) dysregulation and initial oxidative stress; stage B is marked by compensatory angiogenesis and early coronary microvascular dysfunction (CMD) with arteriolar wall thickening; stage C harbors established CMD, capillary rarefaction onset, and vascular barrier disruption driven by ET-1 (endothelin-1) elevation and leukocyte-mediated inflammation; stage D is marked by severe capillary rarefaction and vascular barrier failure with perivascular fibrosis and dominance of antiangiogenic signaling. Angpt2 indicates angiotensin-2; Angptl4, angiotensin-like 4; BH4, tetrahydrobiopterin; CAD, coronary artery disease; EndMT, endothelial-to-mesenchymal transition; eNOS, endothelial NO synthase; FKBP, FK506-binding protein-like; FSTL1, follistatin-like protein 1; HIF-1, hypoxia-inducible factor 1; ICAM-1, intercellular adhesion molecule 1; IL, interleukin; ROS, reactive oxygen species; SMC, smooth muscle cell; TGF, transforming growth factor; TNF-α, tumor necrosis factor-α; VCAM-1, vascular cell adhesion molecule 1; VE-cadherin, vascular endothelial cadherin; and VEGF, vascular endothelial growth factor. Created with BioRender.com.

human HF but have important limitations: MI and thoracic aortic constriction (TAC) surgery impose acute, standardized injuries in young, otherwise healthy animals, which does not reflect the chronic, multicomborbid trajectory of human disease; some reports study a single sex, typically male, despite documented sex differences in vascular and angiogenic responses; and severity of left anterior descending coronary artery ligation or aortic constriction varies across labs, complicating cross-study comparison. Despite these caveats, these models have allowed researchers to study the molecular mechanisms underlying HF development. Notably, cardiac remodeling is accompanied by remodeling of the microvascular bed, and strategies that improve or prevent microvascular damage also improve or prevent cardiac remodeling and dysfunction, suggesting that coronary microvascular remodeling is an integral element of cardiac remodeling, which classically involves cardiac hypertrophy and fibrosis.

HFrEF models include, but are not limited to, MI induced by ligation of the left anterior descending

coronary artery, which promotes tissue damage that varies based on the location of the surgical constriction, triggering cell death, inflammation, scar formation, and loss of contractility, ultimately leading to cardiac dysfunction.¹⁹ The coronary microvasculature also manifests important alterations in this model: ECs and cardiomyocytes undergo cell death rapidly on ischemia induction, promoting hemorrhage and barrier dysfunction, followed by compensatory angiogenesis attempting to repair the damage. However, prolonged ischemia ultimately drives an inflammatory response in ECs and infiltration of immune cells.²⁰ Other models include TAC, which promotes compensatory concentric cardiac hypertrophy followed by dilation,¹⁹ and pharmacological models such as continuous infusion of angiotensin II to induce systemic hypertension and cardiac remodeling, including hypertrophy and fibrosis, over 14 days of administration.¹⁹ In these models, cardiomyocyte hypertrophy and conversion of fibroblasts to myofibroblasts are accompanied by hyperproliferation of VSM cells promoting arteriolar thickening and reduction in capillary density.²¹

Similar microvascular remodeling has been observed in HFpEF models, though with greater difficulty given the syndromic nature and heterogeneity of HFpEF.²² These models include, but are not limited to, genetic models such as db/db mice, ob/ob mice, and ZSF1 rats, or models combining a high-fat diet (HFD) to induce metabolic stress with prohypertensive agents such as angiotensin II or N ω -nitro-L-arginine methyl ester.²³ These models recapitulate diastolic dysfunction without affecting systolic function, alongside metabolic alterations ranging from impaired glycolysis and glucose oxidation to altered fatty acid uptake and oxidation.²²

Coronary remodeling is a consistent feature accompanying cardiac remodeling across distinct etiologies. This convergence points to vascular dysfunction as a shared upstream driver of both microvascular and cardiac remodeling, warranting closer examination of the specific molecular mechanisms linking vascular function with HF progression.

ENDOTHELIUM-DERIVED FACTORS IN HF

Among the endothelial-derived factors discussed to date, NO signaling is the most extensively characterized and evaluated in multiple large clinical trials; EDHF and FSTL1 (follistatin-like protein 1), by contrast, represent comparatively less explored areas, with evidence currently confined to preclinical models and a small number of correlative clinical studies. ECs are situated as the innermost layer of the vascular wall and are in direct contact with the circulation. Due to this strategic location, ECs serve as a key sensor of hemodynamic forces, including blood flow and shear stress, as well as biochemical cues such as inflammatory mediators. In response to these stimuli, ECs release a variety of signaling molecules and regulatory factors that coordinate the activity of other cell types within the cardiovascular system. Their acute response involves regulating vascular tone to control blood flow through the release of vasoactive agents, a process that has been extensively studied and reviewed elsewhere. Here, we summarize recent studies of how endothelial dysregulation contributes to impaired heart function and how interventions may reverse or prevent this damage.

NO, produced by NOS (NO synthases), facilitates vasodilation and inhibits platelet aggregation.²⁴ NOS requires L-arginine, NADPH, and O₂ for NO production.²⁵ In healthy conditions, NO helps maintain vascular homeostasis and regulate myocardial blood flow.²⁶ Adequate NO levels and NOS regulation are pivotal to normal heart function. Several molecular mechanisms converge to regulate NO bioavailability in the failing heart, acting at different points including substrate and cofactor availability, receptor-mediated signaling, and posttranslational modification of NOS itself. Depletion of endothelial NADPH has been reported in the postischemic heart.²⁵ Inhibition of CD38, a receptor with enzymatic

activity that consumes NAD⁺, the precursor for NADPH, in MI models limits NADPH depletion, thereby restoring NO production and vascular relaxation, reducing infarct size, and improving cardiac function.²⁵ GTPCH (GTP cyclohydrolase 1) regulates the production of tetrahydrobiopterin, a crucial NOS cofactor that promotes NOS dimerization and NO biosynthesis.²⁷ EC-specific *Gch1* knockout mice exhibit reduced GTPCH protein and tetrahydrobiopterin levels in the heart under basal conditions, leading to NOS uncoupling, decreased NO production, and increased generation of superoxide and hydrogen peroxide.²⁷ These alterations result in mild induction of genes associated with cardiac hypertrophy and reduction in cardiac output. In the setting of ischemia/reperfusion (I/R) injury, these mice manifest increased infarct size area and elevated left ventricular end-diastolic pressure. Together, these findings indicate that endothelial tetrahydrobiopterin production is essential for maintaining cardiac function and limiting tissue damage in response to cardiac I/R injury.²⁷

NO derived from eNOS also increases NO availability in the central nervous system, thereby inhibiting sympathetic activity.²⁸ HF induced by coronary ligation in rats is associated with reduced eNOS-derived NO in the sympathetic nuclei of the brain, leading to increased sympathetic activity.²⁸ Because heightened sympathetic outflow is a hallmark of HF development, these findings suggest that the cardioprotective effects of NO extend beyond local regulation of vascular tone and include modulation of systemic neurohumoral activation.²⁸

Receptor-mediated signaling also regulates eNOS activity. HF induced by TAC in mice increases S1pr1 (sphingosine-1-phosphate receptor 1) expression in microvascular ECs.²⁹ Endothelial-specific deletion of S1pr1 worsens cardiac dysfunction, hypertrophy, and fibrosis.²⁹ In vitro, ECs overexpressing S1pr1 promote activation of the AKT/eNOS pathway and NO levels. Conditioned medium from these cells reduces fibroblast-to-myofibroblast conversion and limits cardiomyocyte hypertrophy.²⁹ Pharmacological activation of S1pr1 in mice prevents cardiac hypertrophy, fibrosis, and systolic dysfunction.²⁹

Posttranslational modifications can also regulate NO levels. Glutathionylation is a reversible posttranslational modification in which a glutathione molecule forms a covalent bond with a cysteine residue on a target protein.³⁰ S-glutathionylation of eNOS has been shown to reduce NO bioavailability in human aortic ECs exposed to hypoxia-reoxygenation in vitro.³⁰ Conversely, deglutathionylation of eNOS during I/R has been proposed as a mechanism that improves NO availability, enhances vascular function, and reduces cardiac injury.³⁰ EC-specific over-expression of thioredoxin, a cytosolic redox enzyme, decreases eNOS glutathionylation and increases NO bioavailability after I/R, reducing myocardial damage and improving vascular function.³⁰

Clinical studies have reported conflicting changes in circulating nitrites, a measure of NO levels,³¹ with one study reporting a reduction of NO metabolites, or a reduction of NO, in the early stages of HFpEF.³² However, nitrosative stress that reduces NO bioavailability has been observed in patients^{31,33} and in preclinical models.²³ This difference between measures could be due to the method used to evaluate plasma levels of NO or the stage of disease progression. However, therapies that have been studied in clinical trials to increase NO availability have not been successful, such as phosphodiesterase-5 inhibitors,³⁴ isosorbide mononitrate,³⁵ or nitrite supplementation.³⁶

Clinical and preclinical studies have shown that NO regulation is crucial for the comprehensive function of the vascular system, including the coronary vasculature, both arterial and microvascular. Furthermore, NO action in other cardiac cells, such as cardiomyocytes and fibroblasts, suggests its broad relevance and accounts for the large number of studies dedicated to this topic. However, in HFpEF, strategies to increase NO availability have not been successful, unlike in HFrEF.³⁷

Whereas endothelium-derived NO mediates vasodilation primarily in large conduit arteries, EDHFs regulate vascular tone at the microvascular level specifically in small systemic and coronary resistance arteries.³⁸ EDHFs represent a group of EC-derived molecules that hyperpolarize the plasma membrane of adjacent cells, such as VSM cells, thereby reducing their contractility and promoting vasodilation.³⁸

Hydrogen sulfide has been identified as a major EDHF as it induces endothelial and smooth muscle cell hyperpolarization and subsequent relaxation,³⁹ a mechanism linked to protein sulfhydration and hyperactivation of K⁺ channels.³⁹ H₂S is synthesized by CSE (cystathionine γ -lyase). EC-specific CSE knockout mice exhibit enhanced endothelial-to-mesenchymal transition (EndMT), a phenotypic shift from ECs to fibroblast-like cells, increasing perivascular fibrosis, reduced NO bioavailability in the heart, impaired systolic function, and vascular relaxation after TAC. These effects are prevented by endothelial-specific CSE gain-of-function.⁴⁰ This suggests that EDHFs may have more actions beyond mere hyperpolarization and microvascular relaxation. Currently, EDHFs, broadly defined as any molecule with hyperpolarizing activity over vascular cells, are less well studied than other vasoactive agents; however, understanding their existence and mechanisms of action beyond microvascular tone regulation could be particularly beneficial for myocardial dysfunction that occurs with or without CAD.

ET-1 is a potent vasoconstrictor produced by ECs. Its contribution to cardiac remodeling has been demonstrated across multiple HF models, although the strength and mechanism of this contribution vary, as detailed below. Its effects can reach not only the systemic circulation but

also the coronary microvasculature, contributing to myocardial remodeling in HF. Clinical studies have demonstrated a positive correlation between circulating ET-1 levels and several markers of cardiac structural remodeling, including left ventricular mass index, interventricular septal thickness, left ventricular diastolic diameter, atrial dilation, fibrosis, and hypertrophy.^{41,42}

Beyond provoking myocardial damage elicited by vasoconstriction and increased afterload, ET-1 acts directly on cardiac tissue, promoting phenotypic changes via 2 G-protein-coupled receptors: ETA (endothelin receptor type A) and ETB (endothelin receptor type B). Both receptors are expressed in the heart and are differentially regulated under pathological conditions. Notably, activation of ETA has been strongly associated with the development of cardiac hypertrophy.⁴³ The ETA blocker LU135252 prevented angiotensin II-induced cardiac hypertrophy in rats.⁴⁴ However, dual antagonism of ETA and ETB with bosentan⁴⁵ or macitentan⁴⁶ improved cardiac function in rats subjected to aortic constriction or in mice manifesting cardiotoxicity induced by sunitinib.⁴⁶ ET-1 also promotes fibrosis through the induction of EndMT under hyperglycemic conditions.⁴⁷ In early stages, EC-specific deletion of ET-1 prevents the mitochondrial and myofibrillar damage caused by hyperglycemia. At later stages, endothelial ET-1 deletion prevents microvascular rarefaction, reduction in VEGFA levels, and cardiac fibrosis, ultimately improving systolic function. The protective effects observed in endothelial ET-1 knockout mice are mediated, at least in part, by inhibition of TGF- β -induced EndMT, a process initiated by the loss of the cell-to-cell adhesion regulator VE-cadherin (vascular endothelial cadherin [implied]).⁴⁷

Patients with HFpEF can manifest increases in the plasma levels of ET-1.⁴⁸ In HFpEF mice, macitentan has been shown to reduce Mef2a expression and decrease the titin N2B transcript variant, which encodes a shorter and stiffer titin isoform,⁴⁸ indicating that ET-1's effect in HFpEF extends beyond the regulation of vascular tone.

Currently, bosentan and macitentan are approved as pulmonary hypertension treatments. A clinical trial to evaluate macitentan's effect in epicardial artery vasospasm was neutral,⁴⁹ possibly reflecting the episodic and multifactorial nature of coronary vasospasm, which may not be adequately captured by ETA/ETB blockade alone, or owing to limitations in patient selection and end point definition. Whether the ongoing bosentan trial (URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT06432452)⁵⁰ addresses these limitations through refined patient selection, dosing, or end points, remains to be determined. In preclinical HF models, ET-1 manifests a wide range of other effects in the microvascular bed, such as EndMT, promoting perivascular fibrosis, as well as a direct effect on cardiomyocytes, suggesting that the antagonist could have beneficial effects across a wide spectrum of HF-related alterations.

FSTL1 is an angiocrine factor secreted by ECs and VSM cells, essential for maintaining coronary vascular wall homeostasis and protecting the heart from fibrosis and functional decline. EC-specific deletion of *Fstl1*, but not deletion in VSM cells, leads to cardiac enlargement accompanied by increased pulmonary vascular resistance and tricuspid valve regurgitation.⁵¹ In addition, EC-specific *Fstl1* knockout mice exhibit excessive α SMA (smooth muscle actin alpha) expression in the atrial endocardium, heart valves, veins, and microvessels, promoting vascular stiffness.⁵¹ SMAD3 activation is markedly enhanced in these mice, consistent with upregulation of TGF- β signaling in vascular mural cells.⁵¹ Treatment with a TGF- β pathway inhibitor reduces the abnormal α SMA accumulation in the atria and blood vessels of *Fstl1* knockout animals, implicating endothelium-derived FSTL1 as a critical regulator of TGF- β -induced cardiac fibrosis.⁵¹ These results suggest a role of FSTL1 in perivascular fibrosis in the heart, which has also been observed in cardiac remodeling in post-MI patients⁵² and in nonischemic HF.⁵³ Although FSTL1 seems to have an antifibrotic effect in the heart, other studies describe FSTL1 as a profibrotic factor downstream of TGF- β in the lung and skin.⁵⁴ These discrepancies are important to consider, possibly reflecting tissue-specific TGF- β coreceptor expression, differences in the cellular source of FSTL1 (endothelial versus fibroblast versus epithelial), or organ-specific mechanical and inflammatory microenvironments that determine whether FSTL1 signaling promotes or restrains fibrosis. Resolving whether this reflects true biological context-dependence or model-specific artifacts will be important before FSTL1 modulation can be considered a cardiac-selective therapeutic strategy.

VASCULAR OXIDATIVE STRESS

Oxidative stress is a central contributor to endothelial dysfunction across HF contexts, but its drivers differ: in HFpEF-associated models (obesity, diabetes, hypertension), ROS production is driven chiefly by metabolic and hemodynamic stress,⁵⁵ whereas in ischemic and pressure-overload models, ROS arises predominantly from reperfusion injury and neurohormonal activation.^{56,57} Many of its downstream consequences, including NO scavenging, EDHF impairment, and proinflammatory endothelial activation, are addressed in the preceding and following sections. This section therefore focuses specifically on the sources and targeting of ROS themselves, rather than duplicating discussion of this cross-cutting mechanism's downstream effect.

Vascular oxidative stress is one of the primary mechanisms in CMD development. Genetic conditions, aging, diabetes, obesity, and hypertension increase ROS production in vascular cells, disrupting the balance between pro-oxidant and antioxidant mechanisms, affecting

cellular structures through the oxidation of lipids, proteins, and DNA, among other actions. This imbalance leads to oxidative damage in the vasculature and in other tissues, including the heart.⁵⁸ Superoxide anion ($O_2^{\bullet-}$), a major ROS, reacts with NO to form peroxynitrite, reducing NO bioavailability and contributing to endothelial dysfunction.⁵⁹ This dysfunction promotes the release of vasoconstrictors such as thromboxane A_2 , ET-1, and prostaglandin H_2 , and enhances cytokine and chemokine secretion, driving inflammation, vascular remodeling, and further endothelial damage.⁶⁰

All cells within the vascular bed, both systemic and coronary, produce ROS, which makes it difficult to determine exactly where the imbalance between pro-oxidants and antioxidants is occurring. Under mechanical stress, VSM cells generate ROS by LOX (lysyl oxidase), altering elastin structure and increasing arterial stiffness. Also, ROS modulate regulators of Ca^{2+} homeostasis, PKC (protein kinase C), and ROCK (Rho-associated protein kinase), promoting exaggerated contraction.^{58,60} Perivascular adipose tissue contributes to oxidative stress through adipokine and inflammatory mediator secretion.⁶⁰ In the setting of sustained oxidative and inflammatory stress, perivascular adipose tissue shifts to a more proinflammatory phenotype, upregulating leptin, IL (interleukin)-6, and IL-1 β while reducing anti-inflammatory factors such as adiponectin and IL-10.⁶⁰

Studies on oxidative stress focused on the systemic circulation have reported repercussions on cardiac function, and interventions that reduce ROS production contribute to reducing heart damage. Four weeks post-MI in rats, eNOS dimerization is increased in the thoracic aorta but not in the inferior vena cava.⁶¹ In contrast, dimerization of the nNOS (neuronal NOS) isoform is increased in the thoracic aorta but decreased in the vena cava. This pattern corresponds with increased oxidative stress in the vena cava of rats with HF, an effect abolished by 7-NI, an nNOS inhibitor.⁶¹ These findings suggest an important role for nNOS expression and regulation in the vascular stress response.⁶¹ Rat aortic ECs subjected to hypoxia/reoxygenation in vitro manifested increased p38 activation and apoptosis, accompanied by a reduction in DUSP4 (dual-specificity phosphatase 4), a dual-specificity phosphatase that promotes p38 dephosphorylation and inactivation.⁶² This response is exacerbated by DUSP4 knockdown and abolished by treatment with a p38 inhibitor.⁶² EC-specific deletion of DUSP4 in mice subjected to I/R resulted in increased cardiac injury and increased levels of NOX4, a NADPH oxidase family member that generates ROS.⁶² These findings suggest that the regulation of p38 activity and apoptosis in the heart after MI are mediated by oxidative stress in the endothelium.⁶² Mice exposed to hypoxia (10% O_2) for 3 weeks exhibit reduced systolic function and cardiac hypertrophy, an effect exacerbated by VSM

cell-specific deletion of CYB5R3 (cytochrome b₅ reductase 3), an enzyme that maintains sGC (soluble guanylate cyclase) heme iron in its NO-sensitive reduced Fe²⁺ state.⁶³ In addition, a reduction in the relaxation of coronary and pulmonary arteries in response to NO-releasing strategies was observed.⁶³ These findings indicate that the cardioprotective effects of NO depend not only on its proper production in ECs but also on the proper function of the sGC receptor in VSM cells, which requires the heme iron to remain in its reduced state.⁶³

In patients with HFpEF and comorbid chronic kidney disease, proteomic analysis of myocardial tissue has reported alterations in antioxidant enzymes, such as a reduction in GPX1 and an increase in SOD1.⁶⁴ In a preclinical genetic model of diabetes (DM) and chronic kidney disease, heart dysfunction was accompanied by a reduction in the heart of proteins that regulate redox state, such as GPX3 and PRDX1.⁶⁴ In the evaluation of oxidative stress, increased markers of DNA oxidative damage were observed in cardiomyocytes and cardiac ECs.⁶⁴

Oxidative stress is a key mechanism underlying endothelial dysfunction, accompanied by a loss in NO bioavailability, which promotes the inflammatory response driving CMD. As a therapeutic target, antioxidants have been studied in HF; however, their mechanisms of action are not focused solely on vascular or microvascular oxidative stress but act more broadly, such as on polyphenols, coenzyme Q10, or strategies aimed at enhancing the action of endogenous antioxidant enzymes. None has yet been approved for use in patients with HF; however, treatments approved for HF that do not specifically target oxidative stress have shown benefits in this context, such as SGLT2 inhibitors.⁶⁵

INFLAMMATION

The vasculature participates in inflammatory responses, playing roles in vascular permeability, cytokine secretion, and immune cell recruitment. The inflammatory functions of the vascular system contribute to cardiac diseases; conversely, heart diseases associated with an inflammatory response can affect vascular function, promoting oxidative stress, vascular remodeling, and dysfunction. Thus, the impact of vascular inflammation and heart diseases on each other is bidirectional. Just as oxidative stress is a mechanism that occurs both locally in the coronary vascular bed and at the systemic level, the inflammatory response is not exclusive to vascular cells; however, alterations in vascular permeability and the infiltration of immune cells into the tissue are vascular responses characteristic of inflammation, events that occur within the coronary circulation.

Vascular permeability refers to the ability of microvessels to facilitate selective movement of substances between blood and tissues. This process occurs mainly

in postcapillary venules, where a monolayer of ECs supported by pericytes forms a regulated barrier. ECs are connected by junctional complexes such as gap junctions (connexin-based channels that enable intercellular communication), adherens junctions (nectin and VE-cadherin, essential for barrier integrity and cytoskeletal anchoring), and tight junctions (claudins, occludins, JAMs [junctional adhesion molecules], and ESAM [EC adhesion molecule] linked to the cytoskeleton through ZO proteins). The endothelial glycocalyx, composed of proteoglycans and glycosaminoglycans such as heparan sulfate, provides a negatively charged luminal layer that contributes to selective barrier function that is dynamically regulated by ATP, adenosine, bradykinin, histamine, and shear stress, among other mediators.⁶⁶

Inflammatory conditions, such as hypertension, MI, and cardiac transplantation, can promote fluid extravasation in the myocardial interstices by increasing capillary permeability. This increase is driven by glycocalyx degradation, impaired lymphatic drainage, or both, ultimately leading to barrier dysfunction, tissue congestion, and impaired cardiac function.^{67,68} I/R injury models, as well as MI, promote alterations in the function of proteins that regulate EC junctions, along with detachment of pericytes, driving instability of the barrier function.⁶⁹ Systolic dysfunction after MI has been linked to endothelial signaling, and HIF-2 α (hypoxia-inducible factor)-2 α has garnered attention in this context.⁷⁰ Whereas HIF-1 α is consistently described as proangiogenic, the role of HIF-2 α differs according to tissue and condition.⁷⁰ EC-specific deletion of *Epas1/HIF-2 α* results in greater vascular extravasation 12 hours after MI compared with wild-type (WT) MI and sham control hearts.⁷⁰ ECs isolated from HIF-2 α -deficient mice manifest reduced VE-cadherin expression relative to WT mice. Functionally, HIF-2 α deletion leads to impaired systolic function at 7 days and markedly increased cardiac remodeling at 28 days post-MI.⁷⁰ RNA-seq analysis revealed increased expression of *plvap*, a regulator of vascular permeability, and reduced expression of barrier-related genes including *cdh5*, *tjp1*, and *angpt2*.⁷⁰ Moreover, IL-6 and VCAM-1 (vascular cell adhesion molecule 1) expression are increased, indicating enhanced inflammatory activation of the endothelium.⁷⁰ Overexpression of ARNT (also known as HIF-1 β), the heterodimeric partner of HIF-2 α , prevented the barrier dysfunction observed in endothelial HIF2 α -knockout mice.⁷⁰ ARNT also suppressed IL-6 expression by directly binding to its promoter, suggesting a coordinated and cardioprotective endothelial HIF-2 α /ARNT mechanism in MI-induced injury.⁷⁰

Angpt2 (angiopoietin-2), an EC-derived peptide, is elevated in patients with nonischemic HF, and its circulating levels positively correlate with pro-BNP in both ischemic and nonischemic HF.⁷¹ In mice, Angpt2 is highly expressed in ECs in the infarct border zone after MI, downstream of the transcription factor FoxO1 (forkhead

box O1).⁷² This increase promotes pericyte detachment and vascular leakage. EC-specific deletion of Angpt2 prevents these responses and reduces adhesion molecule expression and neutrophil infiltration.⁷² Moreover, EC-specific deletion of Angpt2 reduced levels of heparinase, an enzyme that degrades the glycocalyx component heparan sulfate.⁷² Moreover, sPLA₂ (secretory phospholipase A2), a proinflammatory enzyme linked to increased vascular permeability, is elevated in patients with HF compared with healthy controls and correlates positively with Angpt2 levels.⁷¹

Alterations in the vascular barrier have also been reported in nonischemic preclinical models of HF. In mice subjected to TAC, claudin-5 levels are reduced, accompanied by increased serum albumin in the myocardial interstitium at 4 weeks, with further progression by 8 weeks.⁷³ These findings highlight impaired vascular permeability in a nonischemic preclinical model.⁷³

ECs isolated from db/db mice harboring a loss-of-function mutation in the leptin gene resulting in obesity, insulin resistance, and HFpEF manifest increased VE-cadherin and VEGFA mRNA levels. However, confocal microscopy revealed that the VE-cadherin/CD31 ratio is reduced in the left ventricle and increased in the right ventricle, indicating region-specific alteration of endothelial barrier function in this model of metabolic HF. Transmission electron microscopy further revealed increased spacing between ECs in the apical region of db/db hearts compared with WT mice.⁷⁴ In addition, ECs from db/db mice exhibited reduced levels of VEGF and its receptors relative to WT controls.⁷⁴

A recent study reported that over-expression of Pdzrn3, an E3 ubiquitin ligase, increased leakage of endogenous IgG and fibrinogen around myocardial blood vessels, along with extravasation of red blood cells in the ventricular walls.⁷⁵ Transmission electron microscopy imaging revealed disrupted capillary structure in transgenic mice, accompanied by elevated end-diastolic blood pressure. These findings link PDZRN3 over-expression to vascular barrier disruption, capillary leakage, and diastolic dysfunction.⁷⁵ Although diastolic dysfunction is part of the HFpEF syndrome, the importance of PDZRN3 in disease development warrants further investigation.

Vascular permeability alterations drive peripheral or pulmonary edema, which are cardinal features of HF. Therapies focus on the use of diuretics to relieve the congestion.⁷⁶ Currently, studies have been exploring the mechanisms regulating vascular permeability that could serve as therapeutic targets.⁷⁷ However, in the context of HF, whether regulation of this mechanism could emerge as a therapeutic target remains to be elucidated.

Leukocyte infiltration is a key component of inflammatory responses, and the vascular system, including perivascular adipose tissue, plays a central role by enabling cytokine release and recruitment of immune cells.⁷⁸ Microvascular inflammation induces structural

and functional changes in ECs, including reduced NO production, increased ROS, and heightened inflammatory activation.⁷⁹ This activation leads to the release of chemokines, cytokines, and adhesion molecules, promoting leukocyte adhesion and infiltration into tissues.⁸⁰ VSM cells shift from a contractile to a secretory phenotype under stress, accompanied by cytokine release.⁸¹ Pericytes also secrete TNF- α (tumor necrosis factor- α), IL-1 β , CCL2, and CXCL12 under proinflammatory conditions, further promoting endothelial activation.⁸² Adhesion molecules expressed on the endothelial surface facilitate leukocyte recruitment and can be cleaved and released into the circulation, a characteristic that has been studied as a biomarker in different pathological conditions.^{83,84}

Patients with cardiogenic shock harbor elevated plasma levels of soluble adhesion molecules such as sVCAM-1, sICAM-1, and sE-selectin, which have been proposed as biomarkers of cardiac injury. In addition, plasma levels of sEndocan (also known as ESM1 [EC-specific molecule 1]), a proteoglycan produced exclusively by ECs, are increased and correlate with systolic dysfunction and pro-BNP levels, particularly in conditions such as endocarditis.⁸⁵ Single-nucleus RNA-seq studies of human atrial tissue from patients with heart diseases revealed that those with ischemic heart disease exhibited downregulation of KLF2, a transcription factor that regulates anti-inflammatory genes in ECs.⁸⁶ Pathway analysis in ECs revealed enrichment of lipid accumulation, lipid peroxidation, and inflammatory processes, along with increased expression of CCL2, CCL5, and TNF- α .⁸⁶

During the acute inflammatory phase after MI, ECs upregulate adhesion molecules that recruit circulating immune cells to the infarcted region. During the subsequent reparative phase, these cells release the chemokine CXCL12, which attracts endothelial progenitor cells to promote neovascularization. In patients with ischemia, plasma levels of both ICAM-1 (intercellular adhesion molecule 1) and CXCL12 are elevated.⁸⁷ In post-MI mice, EC-specific ICAM-1 reduction combined with CXCL12 overexpression improved systolic function and cardiac remodeling while reducing leukocyte infiltration.⁸⁷ CXCL12 signals through the receptors CXCR4 and CXCR7,⁸⁸ and CXCR7 is upregulated in injured arteries in both mice and humans.⁸⁸ EC-specific deletion of CXCR7 increases neointima formation and reduces angiogenesis after arterial injury, leading to worsened systolic dysfunction, fibrosis, and infarct size post-MI.⁸⁸ Conversely, CXCR7 gain of function in the heart protects against MI-induced structural and functional alterations.⁸⁸ At 28 days post-MI, endothelial dysfunction, characterized by ROS accumulation, immune cell infiltration in the aorta, and impaired vasorelaxation in response to acetylcholine, is observed.⁸⁹ Depletion of myelomonocytes prevented systolic dysfunction, reduced aortic ROS levels, and restored Acetylcholine-induced vasorelaxation, effects similar to those observed with the angiotensin II receptor

blocker telmisartan.⁸⁹ In ApoE-knockout mice subjected to TAC, Nogo-B, a protein localized to the endoplasmic reticulum, is elevated around atherosclerotic plaques and accompanied by cardiac remodeling and systolic HF.⁹⁰ These effects were prevented in EC-specific ApoE/Nogo-B double-knockout mice. Loss of Nogo-B in ECs reduced levels of mitofusin-2, a protein that maintains endoplasmic reticulum-mitochondria tethering.⁹⁰ This reduction promoted uncoupling of the 2 organelles, limiting endoplasmic reticulum-to-mitochondria Ca^{2+} transfer, decreasing ROS production and VCAM-1 and ICAM-1 expression. Together, these changes mitigated endothelial dysfunction and improved cardiac function.⁹⁰

Patients with HFpEF manifest elevated plasma levels of inflammatory biomarkers that have been correlated with disease severity and prognosis.^{91–94} Patients with HFpEF manifested greater endothelial expression of adhesion molecules, including ICAM-1 and E-selectin, compared with patients with HFrEF, similar to ZSF1 rats, which develop an HFpEF phenotype by 20 weeks of age.⁹⁵ ZSF1 rats manifest diastolic dysfunction, cardiac hypertrophy, and perivascular fibrosis by 21 weeks, without affecting systolic function, compared with lean control rats on a standard diet.⁹⁶ HFD-fed rats also display reduced capillary density, accompanied by capillary dilation, and decreased VE-cadherin levels at 21 weeks, preceded by pericyte loss at 14 weeks.⁹⁶ Notably, pericyte depletion by itself in female mice is sufficient to drive diastolic dysfunction and increase CD45⁺ immune cell infiltration.⁹⁶

Leptin receptor-deficient db/db mice exposed to a 1% salt diet develop diastolic dysfunction, cardiac hypertrophy, and fibrosis after 8 weeks, without changes in systolic function.⁹⁷ Microvascular dysfunction emerged earlier and was associated with reduced cardiac pericyte coverage. After just 4 weeks on the 1% salt diet, these mice also exhibited arteriolar remodeling characterized by increased vessel wall thickness.⁹⁷ RNA-seq analysis in human pericytes, vascular cells providing structural and functional support to ECs, exposed to oxidative stress, manifested upregulation of genes related to inflammation, such as IL1A, IL1B, IL6, and CXCL2.⁹⁶ Also, in conditions of oxidative stress, pericytes increase the expression and release of TNF- α .⁹⁷ Under conditions of oxidative stress, coculture of pericytes with ECs provokes cytokine production by ECs,^{96,97} highlighting the regulatory importance of cell crosstalk in diastolic dysfunction development under inflammatory conditions. Empagliflozin, an SGLT2 inhibitor, reduced the abundance of microvascular inflammatory molecules in patients with HFpEF and in ZSF1 rats, along with reduction of oxidative stress and improvement in vasorelaxation in response to acetylcholine.⁹⁴

Dahl salt-sensitive rats fed a high salt diet develop an HFpEF phenotype.⁹⁸ The treatment with intracoronary infusion of cardiosphere-derived cells, a type of stromal/

stem progenitor cell developed to promote cardiac regeneration, reduced circulating proinflammatory cytokines, fibrosis, and oxidative stress in the aorta.⁹⁸ Single-cell RNA-seq revealed a change in the cardiac distribution of the EC population such that ECs from HFpEF rats treated with cardiosphere-derived cells are more similar to non-HFpEF control rats.⁹⁸

The inflammatory response in the vascular system resulting from an imbalance between proinflammatory and anti-inflammatory signals plays a fundamental role in HF progression. In this context, inflammatory mediators are easily tracked because they are released into the circulation; for this reason, they have been studied not just in preclinical models but also in humans, where they have been correlated with disease severity. Oxidative stress is a precursor to inflammatory activation in the vascular system, and strategies aimed at reducing oxidative stress could be explored, taking advantage of the facile tracking of proinflammatory mediators produced and released by the vascular system.

Taken together, several endothelium-derived molecules identified across these studies represent candidate biomarkers with distinct potential clinical roles: soluble adhesion molecules (sVCAM-1, sICAM-1, and sE-selectin) and sEndocan track acute endothelial activation and correlate with injury severity, making them best suited to early risk stratification after an acute event such as MI or cardiogenic shock; whereas endostatin, FKBPL (FK506-binding protein-like), and neuropilin, discussed below in the context of HFpEF, reflect chronic angiogenic failure and may be more informative for longitudinal monitoring or for identifying patients with HFpEF with a predominantly microvascular phenotype. None of these has yet been validated in prospective trials for treatment selection, but they represent a starting point for biomarker-guided patient stratification in future studies.

The mechanisms described above (NO and EDHF loss, ET-1 excess, oxidative stress, inflammation) do not operate as independent tracks but as a self-reinforcing network (Figure 2). ROS-mediated uncoupling of eNOS reduces NO bioavailability, which removes a key brake on leukocyte adhesion and VSM contraction, thereby promoting inflammatory activation and vasoconstriction. Inflammatory cytokines released by activated endothelium, pericytes, and infiltrating leukocytes further impair eNOS coupling and downregulate proangiogenic signaling, linking inflammation directly to rarefaction. EndMT emerges repeatedly across these pathways, downstream of ET-1/TGF- β signaling, oxidative stress, and loss of endothelial CSE, suggesting it functions as a convergence point where multiple upstream insults produce a common downstream phenotype of perivascular fibrosis and capillary loss. This bidirectional reinforcement among oxidative stress, inflammation, vasoactive imbalance, and rarefaction may help explain why single-pathway interventions (eg,

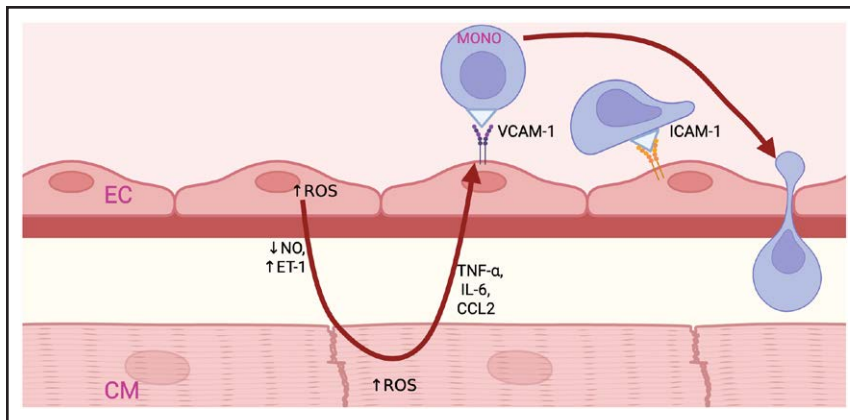


Figure 2. Endothelial cell (EC)–immune cell–cardiomyocyte crosstalk in heart failure (HF).

EC release mediators that affect cardiomyocytes (CM) and vice versa, promoting an inflammatory response with infiltration of immune cells (mono) into the myocardial tissue and release of soluble adhesion molecules (sVCAM-1, sICAM-1) that can be tracked during HF progression. ET-1 indicates endothelin-1; ICAM-1, intercellular adhesion molecule 1; IL, interleukin; ROS, reactive oxygen species; TNF- α , tumor necrosis factor- α ; and VCAM-1, vascular cell adhesion molecule 1. Created with BioRender.com.

NO-restoring therapies) have had limited clinical success: correcting one node may be insufficient when the surrounding network continues to drive dysfunction. This same adaptive-to-maladaptive logic extends to angiogenic regulation, discussed next: the compensatory angiogenic response is progressively undermined by the same oxidative, inflammatory, and EndMT-driving signals described above, and its trajectory is further shaped by metabolic and hormonal inputs.

ANGIOGENESIS AND HF

Angiogenesis is the process of forming new blood vessels from preexisting capillaries or postcapillary venules. It is an organized process that involves multiple phases, including initiation or sprouting, matrix degradation, growth of new vessels, anastomosis between the new vessel and previously formed vessels, and maturation, resulting in the structural and functional completion of the new vessel.⁹⁹ This process initiates in response to inflammatory conditions and is highly regulated by growth factors, cytokines, and multiple signaling pathways.⁹⁹ The opposite process is vascular rarefaction, which is the loss of the number and functionality of microvascular blood vessels, a process observed in certain heart diseases.¹⁰⁰ Rarefaction can be associated with the removal of pro-angiogenic factors, antiangiogenic factors, blood flow disruption, loss of EC and pericyte association, endothelial dysfunction, and EndMT¹⁰⁰ and can contribute to ischemia.¹⁰¹

Angiogenesis is fundamental to heart function. Cardiomyocytes within the healthy heart are surrounded by capillaries, whereas in ischemic cardiomyopathy, capillary density is reduced. Parallel with the phases of compensatory to maladaptive cardiac hypertrophy and ultimately HF, angiogenesis passes through phases, including a compensatory increase in angiogenesis during the initial stress response followed by rarefaction in the maladaptive phase in the failing heart.⁶ The crosstalk between cardiac cells, mediated by factors, cytokines, as well as the activation or inactivation of

signaling pathways, has been the focus of considerable investigation.

Regulators of the Compensatory Angiogenic Phase

During the initial compensatory phase of cardiac remodeling at 14 days following TAC surgery, mice exhibit an increase in microvessel density along with elevated expression of the proangiogenic factors VEGF and angiopoietin.¹⁰² By 28 days post-TAC, however, capillary density and proangiogenic factor expression are reduced, coinciding with impaired systolic function and increased fibrosis.¹⁰² Cardiomyocyte-specific deletion of p38 α MAPK in mice subjected to TAC resulted in exacerbated cardiac dysfunction compared with WT TAC mice.¹⁰³ This was accompanied by more pronounced fibrosis, reduced capillary density, and decreased expression of the proangiogenic factor VEGF during the compensatory phase of hypertrophy. These findings suggest that p38 α MAPK in cardiomyocytes regulates angiogenesis during the compensatory phase by upregulating VEGF.¹⁰³

Despite the known role of numerous proangiogenic factors and signaling molecules in angiogenesis and vascular homeostasis, elucidating the role of their upstream regulators will provide molecular insights into vascular morphogenesis and angiogenesis in fetal and adult life as well as in the failing adult heart. Although elegant work using gene disruption and mutation strategies has uncovered essential roles of diverse transcription factors in vascular development, integrity, and angiogenesis, the ETS (E26 transformation-specific) family of transcription factors has been highlighted as acting at the apex of the vascular program.¹⁰⁴ For example, mice with somatic loss of ETS family transcription factor *Etv2* (also called *Etsrp* [Ets-related protein] 71/ER71) die in utero with complete loss of vasculature.^{105,106} Similarly, knockdown of the ER71 ortholog *Etsrp* in zebrafish also provokes a profound impairment of vasculogenesis,¹⁰⁷ highlighting a central role of *Etv2* in endothelial specification and vascular development in

early development. However, embryonic expression of *Etv2* ceases at mid-gestation,^{105,106,108} suggesting that downstream *Etv2* target(s) in ECs is/are responsible for continuing and facilitating the endothelial program for the remaining gestational period as well as in neonatal and adult life. Indeed, mRNA levels of 2 specific ETS family transcription factors, *Fli1* (friend leukemia integration 1) and *ERG* (ET-related gene), are markedly attenuated in *Etv2*-null embryos.¹⁰⁹ In addition, mice with EC-specific loss of either the *Fli1*¹¹⁰ or *ERG*¹¹¹ gene from gastrulation are viable but manifest impaired vascular morphogenesis and integrity, suggesting that *Fli1* and *ERG* act downstream of *Etv2* and play an independent or cooperative role essential for vascular morphogenesis and homeostasis. In fact, *Fli1* has been identified as being dependent on *Etv2* to initiate early embryonic expression and subsequently acting to direct its own as well as numerous *Etv2* endothelial targets to promote vascular morphogenesis and homeostasis at and beyond mid-gestation, a developmental stage at which the expression of the *Etv2* gene has ceased.¹⁰⁹

In addition, conditional inactivation of both *Fli1* and *ERG* genes in developing or adult endothelium results in impaired vascular integrity and lethality of both developing and adult mice,¹¹² highlighting a cooperative role of *Fli1* and *ERG* in vascular homeostasis and integrity. Importantly, differential gene expression analyses using isolated ECs from WT and EC-specific *Fli1* and *ERG* double-knockout mice revealed marked attenuation of numerous essential endothelial genes, including *Kit*, a receptor tyrosine kinase.¹¹² Intriguingly, a recently reported study has uncovered a signaling axis involving inflammatory cell-derived *Metnl* (Meteorin-like) protein-mediated activation of endothelial c-KIT to promote EC proliferation, angiogenesis, and tissue repair after ischemic injury.¹¹³ On the other hand, accumulating evidence suggests that crosstalk between endothelial *Nrg1* (neuregulin-1) and cardiomyocyte tyrosine kinase receptors *ErbBs* (erythroblastic leukemia viral oncogene homologs) is indispensable in fetal cardiovascular development as well as cardiomyocyte viability and improved contractile activity responses to ischemic injury.^{114–116} Although the emerging new signaling axis seems promising in improving cardiovascular health after stress, the regulatory mechanism(s) governing activation of this signaling axis in the stressed heart is incompletely understood. Currently, it is also unknown whether the *Metnl*-*Kit* signaling axis is in play to repair the damaged human heart. Therefore, future studies will be important to determine whether activation of the *Metnl*-*Kit* axis in stressed heart activates downstream endothelial target(s), such as *Fli1*/*ERG*1 or *HIF-1 α* , which, in turn, activate *Nrg1* gene expression and improve cardiomyocyte viability and contractile activity in ischemic or hypertrophic hearts by activating *ErbB* downstream prosurvival signaling cascades.

Deciphering the underlying mechanism of complex intercellular (immune-EC-cardiomyocyte) communications in stressed hearts will propel identification of novel targets and development of therapeutic strategies with potential clinical benefits.

HIF-1 α is a transcription factor activated under hypoxic conditions whose expression also increases in the early phase of pressure overload and has been established as an essential regulator of angiogenesis during this stage.¹⁰² Cardiomyocyte-specific *HIF-1 α* knockout leads to downregulation of VEGF 3 days after TAC, resulting in reduced contractile function and attenuated cardiac hypertrophy.¹⁰² Collectively, these findings position *Fli1* and *HIF-1 α* as key regulators of vascular morphogenesis and compensatory angiogenesis in the stressed heart.

Transition to the Maladaptive Phase and Capillary Rarefaction

In the maladaptive phase of vascular remodeling, the accumulation of p53 inhibits *HIF-1 α* , suggesting that p53 contributes to capillary rarefaction in the final stage of the vascular remodeling process.¹⁰² *Rdn3*, a Rho-GTPase, has been shown to interact with and stabilize *HIF-1 α* .¹¹⁷ Cardiomyocyte-specific *Rdn3* haploinsufficiency in mice subjected to TAC elicited reduced compensatory capillary formation compared with WT mice, accompanied by decreased protein levels of *HIF-1 α* and VEGFA.¹¹⁷ Conversely, cardiomyocyte overexpression of *Rdn3* prevented cardiac dysfunction and increased capillary density in TAC mice, a response associated with elevated *HIF-1 α* and VEGFA levels in the heart.¹¹⁷

In vitro studies demonstrated that miR-221-3p, which is elevated in myocardial tissue from patients with dilated cardiomyopathy, promotes the binding of *HIF-1 α* mRNA to Ago2 protein, leading to its degradation and a subsequent reduction in *HIF-1 α* protein levels.¹¹⁸ After 4 weeks of TAC (late phase), treating mice with antagomir-221-3p partially prevented systolic dysfunction, cardiac hypertrophy, and fibrosis, while increasing capillary density.¹¹⁸ miR-665 is also increased in patients with HF.¹¹⁹ This microRNA reduces the expression of CD34, a marker of endothelial progenitor cells, and promotes apoptosis and decreases proliferation and tube formation in human umbilical vein ECs.¹¹⁹ Inhibition of miR-665 or overexpression of CD34 in the heart prevented systolic dysfunction, cardiac hypertrophy, and fibrosis, and increased coronary flow reserve, an indicator of myocardial perfusion, in mice subjected to 4 weeks of TAC.¹¹⁹

NO regulates VEGF signaling and recruitment of bone marrow-derived endothelial progenitor cells to sites of injury.¹²⁰ Inhibition of eNOS has been shown to impair vascular regenerative processes.¹²⁰ Dietary supplementation with nitrites in mice limits this impairment.¹²⁰ The administration of empagliflozin, an SGLT2

inhibitor currently approved for HF, in TAC-exposed mice increases endothelial NO levels through activation of the AKT/eNOS pathway.¹²¹ This response is associated with enhanced cardiac vascular density and improved cardiac dysfunction.¹²¹

Metabolic and Hormonal Modulators of Angiogenesis

Beyond their well-documented effects on angiogenic signaling, these same metabolic and hormonal factors also intersect with the oxidative stress and inflammatory pathways discussed earlier; for instance, leptin promotes endothelial IL-6 and IL-1 β release independent of its angiogenic actions, and hyperglycemia directly increases endothelial ROS production. The angiogenesis-specific effects reviewed below should therefore be interpreted as one arm of a broader metabolic-vascular interaction rather than the entire process. Pathways associated with metabolic dysfunction have been investigated in the context of HF and angiogenesis. LepR (Leptin receptor) signaling promotes EC proliferation and migration as well as angiogenesis in the heart.¹²² EC-specific deletion of LepRs in TAC-exposed mice blunts cardiac hypertrophy and cardiac dysfunction after 8 and 20 weeks of pressure overload.¹²² Moreover, LepR deletion reduces activation of signaling pathways downstream of the receptor, including ERK1/2 and AKT/mTOR (mammalian target of rapamycin), leading to increased autophagy (a process known to participate in angiogenesis) and enhanced EC sprouting.¹²²

Elevated glucose levels can impair endothelial sprouting and maturation during angiogenesis. A transgenic pig harboring a mutation in the insulin gene causing neonatal diabetes exhibits marked cardiac hypertrophy accompanied by a reduction in pericytes surrounding blood vessels, systolic dysfunction, and fibrosis.¹²³ These abnormalities are exacerbated under ischemic conditions. Treatment of these diabetic pigs with T β 4 (thymosin β 4), a proangiogenic factor, enhances capillary formation and pericyte coverage, thereby improving cardiac function.¹²³ In db/db mice, a reduction in capillary density has been observed across different regions of the heart. Although mRNA levels of VEGFs and their receptors are increased in ECs, in situ levels of VEGFR2 (VEGF receptor 2) and VE-cadherin are reduced throughout the myocardium.⁷⁴

DPP-4 (dipeptidyl peptidase-4), a proteolytic enzyme, is upregulated in HFD-exposed mice and contributes to impaired angiogenesis by inhibiting the FGF-2/EGR1/VEGFA pathway in fibroblasts.¹²⁴ DNA microarray analysis of HFD-exposed mice treated with linagliptin, a DPP-4 inhibitor approved for type 2 diabetes, for 2 weeks revealed enrichment of KEGG terms related to the HIF-1 α signaling pathway and angiogenesis.¹²⁴ Furthermore, linagliptin treatment prevented systolic dysfunction and preserved capillary density in HFD mice.¹²⁴

In diabetic cardiomyopathy induced by streptozotocin, levels of miR-195 are elevated and correlate with cardiac dysfunction.¹²⁵ Treatment with anti-miR-195 blunts cardiac dysfunction and increases capillary density in the heart. This effect appears to be mediated directly through ECs, as in vitro overexpression of miR-195 reduces tube length and connectivity in endothelial tube-formation assays.¹²⁵

Hormonal regulation of angiogenesis in rats has been investigated in right ventricular (RV) failure in the pulmonary artery banding model.¹²⁶ Female rats were protected from elevated systolic pressure, systolic dysfunction, and RV remodeling compared with males. This protection is associated with upregulation of SphK1 (sphingosine kinase 1), a proangiogenic kinase. In vitro studies further demonstrated that estrogen (E2) promotes tube formation in ECs isolated from the female RV ECs, but not in male RV ECs. Notably, the SphK1 inhibitor PF-543 abolished the E2-induced angiogenic response in female RV ECs.¹²⁶



Temporal Phases of Post-MI Angiogenesis

Angiogenesis post-MI supports tissue regeneration and reoxygenation of the damaged myocardium. Bulk RNA-seq analyses have enabled the identification of distinct temporal patterns of gene upregulation and downregulation, allowing the angiogenic response to be divided into phases.¹²⁷ In the early phase (2 days post-MI), there is a marked upregulation of angiogenesis-related genes, such as Angpt2, which return to baseline levels by days 7 and 28. In the second phase, corresponding to the scar-formation stage (7 days post-MI), members of the Rho-GTPase family are upregulated together with components of the robo-slit, semaphorin-plexin-neuropilin, Wnt, MAPK, and Hedgehog signaling pathways.¹²⁷ These pathways are associated with EC migration, angiogenesis, and blood vessel assembly. In the late phase (28 days after MI), ECs remain active and do not fully return to a quiescent state. Instead, they display a mixed pattern of up- and downregulation of pathways involved in inflammation and proliferation.¹²⁷

Acute MI patients exhibit increased levels of CYP2J2 (cytochrome P450 2J2), which is expressed predominantly in ECs, smooth muscle cells, and cardiomyocytes, and metabolizes arachidonic acid to generate cardioprotective epoxyeicosatrienoic acids.¹²⁸ In mice, cardiomyocyte-specific overexpression of CYP2J2 partially prevents cardiac dysfunction 8 weeks after MI, reduces fibrosis, and decreases cell death in the border and remote zones of the infarct.¹²⁸ Moreover, CYP2J2 overexpression increased the levels of the proangiogenic factors VEGF and FGF in the heart, as well as the number of vessels in the border and remote zones, although not in the infarct core.¹²⁸ In vitro, treatment of human umbilical vein ECs with epoxyeicosatrienoic acids under

both normoxic and hypoxic conditions promotes tube formation, which is associated with increased expression of HIF-1 α , VEGF, and FGF.¹²⁸ These effects are abolished by an epoxyeicosatrienoic acid antagonist. The protective effects of epoxyeicosatrienoic acids are mediated through activation of the Jagged1/Notch1 signaling pathway.¹²⁸

Fibroblast differentiation into myofibroblasts, which also release proangiogenic factors,¹²⁹ precedes endothelial proliferation after MI in mice.¹³⁰ Jph2 (juxtaphilin-2), a protein expressed exclusively in human and mouse cardiac fibroblasts, plays an essential role in this process.¹³¹ Cardiac fibroblast-specific deletion of Jph2 impairs angiogenesis after MI, leading to reduced mRNA levels of Vegfa, Vegfc, and Vegfd, as well as decreased numbers of CD31⁺ ECs and SMA⁺ VSM cells in the infarct and border zones, but not in the remote zone.¹³¹

Neutrophil infiltration in the infarct and border zones, as well as into the circulation 4 weeks after MI, is associated with elevated expression of CXCL2 and its receptor CXCR.¹³² In a mouse model of MI, the experimental approach of treatment with an anti-CXCL2 antibody mitigates cardiac dysfunction and adverse remodeling while also improving angiogenesis.¹³² The scavenger receptor for CXCL12, CXCR7, is upregulated in ECs from arteries subjected to various injuries in both humans and mice.⁸⁸ In human umbilical vein ECs, CXCR7 blockade reduces tube formation, and EC-specific deletion of CXCR7 impairs revascularization in a hindlimb ischemia model.⁸⁸ In a mouse MI model, endothelial-specific CXCR7 deletion worsens cardiac dysfunction, increases infarct size and fibrosis, and reduces vascularization in the scar area.⁸⁸ Conversely, CXCR7 gain-of-function after MI decreases infarct size.⁸⁸

ROS, produced primarily by mitochondria in ECs, are normally balanced by antioxidant enzymes, such as MnSOD. EC-specific overexpression of MnSOD after MI improves cardiac function, reduces infarct area, and increases capillary density in the infarct region.¹³³ These effects occur through activation of the AKT and ERK1/2 pathways, highlighting the role of ROS signaling in the regulation of angiogenesis.¹³³ On the other hand, IGFBP4 (insulin-like growth factor-binding protein 4) has been shown to reduce DNA damage after MI, thereby decreasing fibrosis and increasing the protein levels of the proangiogenic factor Angpt1.¹³⁴ Dapagliflozin, an SGLT2 inhibitor, protects against infarct damage 4 weeks after MI, blunting cardiac hypertrophy and fibrosis, improving cardiac function, and enhancing angiogenesis during the early compensatory phase.¹³⁵

The proangiogenic effects of VEGFB in MI depend on whether its release occurs through paracrine or autocrine mechanisms. Paracrine VEGFB derived from cardiomyocytes promotes the formation of new capillaries, whereas autocrine VEGFB from ECs contributes only minimally to neovascularization.¹³⁶ SUMOylation, a

posttranslational modification involving covalent attachment of SUMO molecules to lysine residues of target proteins, regulates their localization, stability, and function. SUMO1, a key SUMOylation enzyme, is increased in the infarct and border zones after MI.¹³⁷ EC-specific SUMO1 deletion alters multiple cellular populations in the infarcted heart and promotes a proliferative, proangiogenic endothelial phenotype. In vitro, SUMO1 knockdown in human umbilical vein ECs enhances endothelial proliferation and tube formation in response to VEGFA, whereas SUMO1 overexpression reduces cell proliferation and tube-forming capacity.¹³⁷

Despite the strength of this preclinical evidence, therapeutic angiogenesis has had limited clinical success: trials of VEGF and FGF gene or protein therapy for ischemic heart disease largely failed to improve perfusion or outcomes despite robust preclinical efficacy.¹³⁸ Proposed explanations include redundancy among angiogenic pathways in vivo, a mismatch between short-term rodent injury models and chronic human coronary disease, insufficient control over the dose, duration, and location of factor delivery, and patient comorbidities (age, diabetes, atherosclerosis) that blunt angiogenic responsiveness in ways preclinical models rarely capture. This gap underscores the need for translational models that better reflect the chronicity and heterogeneity of human diseases.

Mapped onto the American College of Cardiology/American Heart Association staging framework, the early compensatory angiogenic response described above is most consistent with stage B, in which structural changes are present but systolic/diastolic function may still be adequately compensated. The subsequent decline in proangiogenic signaling and capillary rarefaction coincides with the transition to stage C, in which structural changes are accompanied by overt functional impairment and symptoms. Persistent, severe rarefaction and vascular barrier failure, discussed above in the context of end-stage remodeling, are more characteristic of stage D.

Angiogenic Failure in HFpEF

Angiogenesis has been less extensively studied in HFpEF; however, HFpEF is closely associated with chronic low-level inflammation that contributes to microvascular rarefaction.^{23,101} Biomarker studies comparing patients with HFpEF and HFrEF have revealed that levels of VEGFR, an angiogenic marker, are significantly lower in HFpEF.¹³⁹ Neuropilin levels are also reduced and have been reported as predictors of rehospitalization in patients with HFpEF.¹³⁹ In addition, the antiangiogenic secreted proteins endostatin and FKBPL are elevated in the plasma of patients with HFpEF.^{140,141} Postmortem analyses of cardiac tissue from patients with HFpEF revealed the presence of CAD, vascular stenosis, and

reduced microvascular density, which negatively correlated with the degree of fibrosis.¹⁰¹

A comprehensive single-cell RNA-seq study in a mouse model of HFpEF further elucidated mechanisms associated with the development of the syndrome. Cardiac fibroblasts exhibit increased expression of the anti-angiogenic gene *Angptl4* and downregulation of at least 8 proangiogenic genes, highlighting a significant role for fibroblasts in regulating endothelial behavior and vascular growth.¹⁴² In ECs, genes involved in angiogenesis, including pathways related to migration, sprouting, and VEGF signaling, are predominantly downregulated.¹⁴² Db/db mice, which develop diastolic dysfunction at 6 months of age, manifest a reduction in capillary density accompanied by a decrease in pericyte number.⁹⁷ Single-cell RNA-seq analysis revealed reduced EC clusters associated with angiogenesis and proliferation.⁹⁷ In vitro studies suggest that pericytes induce EC cycle arrest through a mechanism involving pericyte-derived TNF- α and downregulation of Cyclin 1 in ECs.⁹⁷

Inflammatory pathways have also been implicated in the regulation of angiogenesis in HFpEF. Deletion of ESAM, an EC-selective adhesion molecule involved in immune and inflammatory responses, leads to cardiac hypertrophy and diastolic dysfunction without affecting systolic function, and promotes microvascular rarefaction.¹⁴³

CONCLUSIONS

Across the pathways reviewed here, the robustness and maturity of evidence vary substantially. NO signaling and, to a lesser extent, ET-1 antagonism are supported by large clinical trials, although these trials have largely shown limited efficacy. Oxidative stress and inflammation are well-established contributors to endothelial dysfunction, backed by convergent preclinical and correlative clinical data, but have not been tested as pathway-specific clinical trial targets. EDHF signaling, FSTL1 modulation, and several angiogenic and metabolic modulators (VEGF, FKBPL, neuropilin, the leptin/DPP-4 axis) remain early-stage, supported chiefly by preclinical models or single-cohort clinical correlations. Recognizing this hierarchy matters for prioritization: the NO/ET-1 axes, despite past trial failures, may warrant refined trial design given their strong mechanistic grounding, whereas EDHF, FSTL1, and most angiogenic/metabolic modulators require further mechanistic and correlative validation before clinical testing is justified.

Considered in the context of American College of Cardiology/American Heart Association staging, the mechanisms reviewed here cluster loosely by disease phase: oxidative stress and early NO/EDHF dysregulation are detectable from Stage A/B onward, driven largely by risk factors and early structural change; ET-1 elevation and inflammatory activation intensify with the transition

to symptomatic stage C; and angiogenic failure, capillary rarefaction, and vascular barrier dysfunction become most pronounced in advanced stage C/d disease (Figure 1). This staging is necessarily approximate, as few of the studies reviewed here directly stratified patients by American College of Cardiology/American Heart Association stage rather than by HFrEF/HFpEF classification or model type.

Proper function of the vascular system contributes to cardiac performance through the hemodynamic coupling between the heart and the vasculature, a concept that has been comprehensively reviewed by others.^{3,4} Here, we approach this association from a molecular overview, framing vascular dysfunction in HF as following a common adaptive-to-maladaptive arc. Preclinical models of both systolic and diastolic dysfunction have revealed the mechanistic involvement of both systemic and coronary vascular cells as drivers of adverse cardiac remodeling and functional impairment, spanning endothelial signaling, oxidative stress, inflammation, and angiogenic remodeling as interconnected rather than independent processes.

Elucidating these mechanisms represents an essential step toward the development of therapeutic strategies focusing on vascular biology in HF. Restoring endothelial NO signaling and targeting the microvascular ET-1 axis remain conceptually attractive, and interventions on inflammation, vascular barrier dysfunction, and cell-specific signaling pathways continue to hold translational potential; however, mechanistic promise at the preclinical level has not consistently translated into clinical benefit; NO-restoring therapies and ET-1 antagonism have both shown limited or no efficacy in clinical trials, and angiogenic gene/protein therapies have largely failed despite robust preclinical rationale. Among the pathways reviewed, EDHF/H₂S signaling and FSTL1 modulation may represent more tractable near-term targets than NO restoration, given that they act through mechanisms distinct from the repeatedly unsuccessful NO-donor and PDE5-inhibitor approaches and have not yet been tested in large clinical trials; similarly, biomarker-guided stratification, for example, distinguishing patients with predominantly inflammatory versus angiogenic-failure phenotypes, could help identify subgroups more likely to benefit from pathway-specific interventions such as ET-1 antagonism or anti-inflammatory therapy. This recurring gap likely reflects the limits of single-pathway interventions tested in relatively homogeneous, acute-injury animal models against the chronic, multifactorial, and patient-heterogeneous nature of human HF.

Finally, prioritizing vascular health, microvascular stability, and vascular-cardiac communication through combinatorial or network-level approaches and more translationally faithful preclinical models represents a promising strategy to complement existing neurohumoral therapies and address unmet needs in HF.

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